

USA, Turkey

WMO: 171880

Lat: 38.683N

Lon: 29.400E

Elev: 3015

StdP: 13.16

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.3	21.9	9.0	9.7	24.2	12.6	11.7	27.0	20.3	39.7	17.5	37.5	3.4	280	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.7	92.9	64.0	90.3	63.2	87.9	62.9	67.5	86.7	66.0	84.5	64.7	82.7	6.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	88.7	75.3	59.2	84.0	73.6	57.9	80.0	72.8	33.7	86.8	32.4	84.5	31.4	82.7	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.0	11.7	DB	12.8	96.7	6.2	3.1	8.3	98.9	4.7	100.7	1.2	102.5	-3.3	104.7	(4)
(5)			WB	11.7	69.6	5.8	2.4	7.6	71.4	4.2	72.8	0.9	74.2	-3.3	75.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.3	37.2	39.5	44.5	51.7	60.4	68.0	74.1	74.7	66.5	57.5	47.8	40.2		
	DBStd	14.41	7.20	7.04	6.54	6.65	5.92	5.49	4.47	4.30	5.54	6.29	6.65	6.67		
	HDD50	1374	398	295	192	55	2	0	0	0	0	13	112	307		
	HDD65	4379	863	713	635	400	164	31	0	1	47	242	515	768		
	CDD50	3300	0	2	21	107	326	540	747	766	495	246	47	3		
	CDD65	832	0	0	0	1	23	121	282	303	92	10	0	0		
	CDH74	9787	0	0	1	33	345	1417	3340	3322	1181	148	0	0		
	CDH80	3804	0	0	0	1	52	450	1447	1440	393	21	0	0		
	WSAvg	4.4	4.1	4.6	4.6	4.3	4.3	4.6	5.1	5.2	4.3	4.1	3.5	3.8		
	PrecAvg	20.3	2.8	2.6	2.3	2.2	1.9	1.0	0.6	0.4	0.8	1.8	2.3	3.3		
(15) Precipitation	PrecMax	29.5	8.3	6.7	4.9	4.8	4.8	5.0	3.0	1.2	4.6	5.6	6.0	8.3		
	PrecMin	13.6	0.1	0.3	0.4	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.3		
	PrecStd	4.0	1.9	1.4	1.2	1.1	1.2	0.9	0.7	0.4	0.9	1.2	1.4	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.5	63.3	69.6	78.7	84.6	91.6	96.9	95.8	91.8	82.8	70.8	61.2		
		MCWB	45.9	46.8	51.0	55.6	60.8	63.4	64.9	64.7	60.7	59.1	53.4	46.4		
	2%	DB	52.7	58.2	64.4	73.4	81.2	88.0	93.3	92.9	88.2	78.2	66.7	55.6		
		MCWB	44.4	45.8	48.5	54.2	59.2	62.8	65.0	63.7	60.5	57.2	51.4	45.9		
	5%	DB	50.0	54.2	60.7	69.4	78.1	85.1	90.5	90.3	84.6	74.2	63.1	52.6		
		MCWB	43.4	44.4	47.2	52.8	58.5	62.0	63.9	63.2	60.2	55.9	50.0	45.2		
	10%	DB	47.3	50.7	57.1	65.4	74.7	82.1	87.7	87.7	81.0	70.6	59.5	49.8		
		MCWB	41.9	43.2	45.9	51.3	57.4	61.3	63.3	63.1	59.3	54.6	49.1	44.5		
	0.4%	WB	47.9	49.8	52.8	57.3	63.4	66.9	70.0	69.1	65.6	62.2	56.4	49.9		
		MCDB	53.1	59.8	66.8	73.5	80.1	84.8	88.9	88.8	85.1	79.0	65.7	55.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.6	46.8	50.1	55.5	61.2	64.7	68.0	67.3	63.6	59.1	53.6	47.9		
		MCDB	51.0	55.4	60.9	70.9	77.1	82.4	87.7	86.1	81.2	72.0	62.5	53.0		
	5%	WB	44.0	45.2	48.2	53.7	59.6	63.4	66.4	65.9	62.1	57.5	51.9	46.5		
		MCDB	48.8	52.8	58.3	67.2	74.8	80.7	85.3	84.2	78.3	69.7	59.7	51.1		
	10%	WB	42.5	43.7	46.6	52.0	58.1	62.2	64.9	64.7	60.7	56.1	50.1	45.0		
		MCDB	46.6	50.0	55.6	63.5	72.1	78.9	83.0	82.6	76.1	67.6	57.3	49.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	15.7	18.1	19.9	22.4	23.9	25.5	24.7	24.7	21.4	18.8	14.7		
		MCDBR	17.9	23.0	25.7	27.9	27.8	27.3	28.9	28.0	31.1	27.8	25.8	19.9		
		MCWBR	11.2	13.0	13.1	12.8	11.5	9.7	9.6	8.9	11.6	12.4	13.8	12.1		
	5% WB	MCDBR	15.4	19.5	21.4	24.9	25.3	25.4	25.0	24.5	25.1	22.7	19.4	14.5		
		MCWBR	10.0	11.5	11.4	11.8	11.1	10.0	9.4	8.8	10.3	10.7	10.5	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.327	0.366	0.397	0.382	0.357	0.352	0.354	0.341	0.338	0.312	0.300		
	taud		2.451	2.377	2.274	2.212	2.312	2.400	2.408	2.384	2.421	2.432	2.479	2.491		
	Ebn at Noon		280	284	282	279	285	291	292	289	287	277	272	271		
	Edh at Noon		26	32	39	44	41	37	37	37	33	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		665	912	1291	1665	2005	2312	2426	2136	1707	1203	838	610		
	RadStd		81	124	113	151	137	158	75	82	111	71	82	82		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page